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The effect of pharmacological and non-pharmacological interventions on pain control after orthodontic treatment: a systematic review and network meta-analysis

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Abstract

Background Pain is a common side effect of orthodontic treatment, affecting both patient experience and treatment outcomes. At present, a combination of pharmacological and non-pharmacological therapies is used to alleviate post-orthodontic pain. This study aimed to compare and rank the effectiveness of pharmacological and non-pharmacological interventions for post-orthodontic pain management using a Bayesian network meta-analysis.

Methods Following PRISMA 2020 guidelines, we searched PubMed, Embase, Web of Science, and the Cochrane Library up to April 2025 for randomized controlled trials evaluating pharmacological or non-pharmacological interventions for orthodontic pain relief. Studies were included if they assessed pain using the Visual Analog Scale (VAS) at 24–48 h. Two reviewers independently screened studies and extracted data, and risk of bias was assessed using the Cochrane RoB 2.0 tool. A Bayesian network meta-analysis was performed to estimate weighted mean differences and rank interventions using SUCRA values.

Result A total of 37 RCTs involving 2,430 participants and 16 intervention arms were included. Pharmacological therapies showed greater efficacy than non-pharmacological methods. Etoricoxib and naproxen ranked highest for pain reduction at both 24 h and 48 h, while structured telephone calls, vibration therapy, and low-level laser therapy demonstrated moderate benefits.

Discussion Pharmacological interventions—particularly etoricoxib and naproxen—were associated with the greatest pain reduction following orthodontic treatment. Structured behavioral and physical approaches may provide additional benefits. Study limitations included variations in intervention parameters and study quality, which may limit the certainty of evidence.

Keywords Orthodontic pain, Etoricoxib, Low-level laser therapy (LLLT), Bayesian

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Introduction

Externally employed forces are utilized for orthodontic tooth movement, initializing tooth displacement within the periodontal ligament space and creating areas of pressure and tension. The alveolar bone tissue reacts to the resulting stimuli and remodels itself through bone resorption and deposition, ultimately completing tooth rearrangement [1]. The corresponding teeth alteration stimulates nerve endings in the periodontal ligament. During the process, various mediators, including substance P, prostaglandins, bradykinin and serotonin, are released, which are believed to contribute to the physiological responses perceived as pain [2].

Orthodontic treatment often induces pain [3].It has been reported that approximately 90%–95% of patients experience varying degrees of discomfort or soreness following fixed appliance treatment, with additional pain often occurring after monthly adjustment visits during the extended treatment period. Clinical observations suggest that the peak intensity of post-orthodontic pain commonly occurs within 24–48 h after appliance activation [4].

Pain diminishes patient satisfaction with treatment effectiveness and undermines adherence to orthodontic protocols. Additionally, it causes a considerable number of patients to either halt or resist treatment, thereby potentially compromising expected therapeutic outcomes [5].

In summary, pain exerts multifaceted adverse effects throughout orthodontic treatment. Consequently, mitigating treatment-induced pain represents a clinically imperative issue for both orthodontists and patients. Moreover, a variety of pain management strategies have been proposed to broaden clinicians' options for optimizing patient care. However, the selection of an optimal pain-control approach remains controversial in routine clinical practice.

Notably, current research on orthodontic pain control primarily focuses on pharmacological and non-pharmacological interventions. Pharmacological interventions primarily consist of nonsteroidal anti-inflammatory drugs (NSAIDs), such as ibuprofen and etoricoxib, which are commonly prescribed to reduce pain through inhibition of inflammatory mediator synthesis [6].Additionally, medications including acetaminophen and aspirin have also been investigated for managing post-orthodontic pain.

In recent years, there has been a significant increase in research focused on non-pharmacological management, complementing pharmacological approaches. For instance, low-level laser therapy (LLLT) has been proposed to exert anti-inflammatory and neuroregenerative effects by delivering focused energy, and has been investigated in various dental disciplines [7].Chewing therapy (chewing gum or masticatory devices) is hypothesized to reduce orthodontic pain by decreasing ischemia through cyclic compression and decompression of the periodontal ligament during mastication [8].Transcutaneous acupoint electrical stimulation (TAES) has been explored as a method for pain modulation via surface electrodes applied over specific acupoints or along relevant neural pathways [9].Low-intensity high-frequency vibration therapy has been suggested to influence cellular responses within periodontal tissues, including osteocyte-derived signaling pathways related to osteoclast activity [10].Additionally, verbal behavior modification (VBM) aims to influence pain perception through psychological cues delivered via phone calls or text messages [11].

Most studies have suggested that both pharmacological and non-pharmacological interventions may provide measurable pain relief following orthodontic procedures. However, it is important to highlight that there is currently no consensus regarding optimal dosing regimens for pharmacological agents or standardized clinical protocols for non-pharmacological interventions in orthodontic pain management. In summary, pain management during orthodontic treatment remains a key clinical concern. Although numerous randomized controlled trials have examined individual interventions, their comparative clinical effectiveness has not been systematically established.

Therefore, the objective of this study was to compare, using a Bayesian network meta-analysis, the effectiveness of various pharmacological and non-pharmacological interventions in reducing pain intensity during the initial phase of orthodontic treatment. This phase is universally characterized by procedures that induce significant discomfort, primarily including separator placement, initial archwire insertion in fixed appliances, and the initial placement of clear aligners (invisible aligners). By encompassing these key initial interventions, this NMA aims to provide a comprehensive evidence base for pain management at the onset of orthodontic therapy.

Methods

The systematic review and NMA has been registered (registration number: CRD420251107001) with the International Prospective Register of Systematic Reviews (PROSPERO). The study was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement [12].

Search strategy

This study rigorously adhered to the PRISMA-NMA guidelines for conducting and reporting [13]. A systematic search was conducted across PubMed, Embase, the

Cochrane Library, and Web of Science. Crucially, intervention terms were deliberately excluded from search queries. Instead, we identified studies featuring "pain" OR "discomfort" in titles/abstracts. No restrictions were imposed regarding participants' age, gender, or ethnicity. Only studies published in English were included, while conference abstracts, case reports, and non-peer-reviewed articles were excluded. Retrieved records underwent secondary screening to confirm utilization of orthodontic pain-relieving interventions. The investigation covered the entire timeline of each database from inception to April 2025. The search strategy for the PubMed, EMBASE and Cochrane database is presented in Supplementary Tables 1 to 4.

Inclusion criteria and exclusion criteria

Study eligibility was defined using the PICOS framework (Population (P): Orthodontic Patients undergoing initial pain-inducing orthodontic procedures, including but not limited to separator placement, initial archwire insertion with fixed appliances, or the first/initial activation of clear aligners.; Intervention (I): Acetaminophen-post, Acetaminophen-pre, Acetaminophen-pre-post, Aspirin-pre-post, Attention call, Chewing, Etoricoxib-pre-post, Ibuprofen-post, Ibuprofen-pre, Ibuprofen-pre-post, LLLT, Naproxen-pre, Naproxen-pre-post, Structured call, TEAS, Vibration; Control (C): Pleacbo, Outcome (O):24 H and 48 H VAS; Study (S): All relevant randomized controlled trials were included). Included were prospective randomized controlled trials (RCTs) that evaluated any pharmacological or non-pharmacological intervention aimed at mitigating orthodontic pain, with no minimum sample size requirement. Control groups received routine care without additional interventions, with outcomes assessed via VAS scores at 24–48 h post-intervention.

Excluded were publications of non-randomized trials, in vitro or animal studies, and investigations involving non-treatment-naïve orthodontic patients. Studies that failed to report VAS scores at 24–48 h, those lacking active interventions for post-procedural pain management, and papers with unavailable extractable data were also excluded.

Data collection

All studies underwent dual independent dual screening by two authors (X.D and H.Q). During the identification phase, automation tools were applied to remove duplicate records and to preliminarily exclude clearly ineligible records based on predefined criteria. After removing duplicates, titles and abstracts were assessed for preliminary eligibility with the assistance of automation tools, supplemented by independent manual review. Full-text articles were then evaluated for final inclusion. Discrepancies were resolved through discussion or adjudication by a third reviewer, with contentious articles re-examined until consensus was reached. Data extraction employed a pretested standardized form capturing:

1. Study identification: first author and publication year;
2. Methodological design;
3. Participant characteristics: sample size, mean age;
4. Intervention protocols: pharmacological/non-pharmacological details (dose, frequency, administration route/timing);
5. Orthodontic procedures;
6. Comparator assessment;
7. Outcome metrics (24–48 h VAS scores)

Outcomes statistical analysis

A Bayesian framework was implemented for NMA of mixed treatment comparisons. Using the gemtc package in R, random-effects NMA was performed via Markov chain Monte Carlo (MCMC) methods with 50,000 iterations. The analysis comprised three sequential components: First, a consistency model estimated all relative effects under consistency constraints, with results expressed as weighted mean differences (WMD) and 95% credible intervals (CrI). Treatment rankings were generated via rankograms, while interventions were ordered by surface under the cumulative ranking curve (SUCRA) for cross-outcome comparison.

Confidence in the network meta-analysis estimates was assessed using the GRADE approach adapted for network meta-analysis (GRADE-NMA) [14].This framework was applied to evaluate the certainty of evidence for each comparison based on predefined methodological domains.

Where reported, subgroup analyses were planned to explore potential differences in pain perception by age group and gender, as these factors may influence analgesic response. However, such analyses were limited by incomplete reporting across studies [15].Subgroup analyses by orthodontic procedure (e.g., separator placement, archwire insertion, aligner activation) were initially planned. However, due to insufficient data reporting or variability in study protocols, it was not feasible to perform these subgroup analyses in the main analysis. Where possible, a sensitivity analysis was conducted to explore the impact of different orthodontic procedures on pain outcomes. This was done by differentiating major categories where the data allowed. Due to limited reporting across studies, results from subgroup and sensitivity analyses should be interpreted as exploratory.

To assess the robustness of the findings, sensitivity analyses were conducted by excluding studies judged to have a high risk of bias and, where applicable, by

sequentially excluding individual studies from the network. The sensitivity analyses showed that the direction and magnitude of the pooled effect estimates remained largely unchanged, and no clinically meaningful differences were observed. In addition, treatment rankings based on SUCRA values exhibited minimal variation, with the leading interventions remaining consistent with those identified in the primary analysis. Owing to data limitations, results from subgroup- and procedure-related sensitivity analyses were interpreted as exploratory.

Subsequently, inconsistency analysis employing node-splitting models verified trial consistency within the network, with between-study heterogeneity quantified via I^2 statistics.Finally, this study, being a systematic review that did not involve human subjects or identifiable private data, was exempt from the requirement of institutional review board approval. SUCRA curves and funnel plots were generated using STATA 17.0.

Result

Literature screen results

A total of 2560 pieces of literature were retrieved. Following a preliminary screening by title and abstract, the literature was rescreened by reading the full text, resulting in the inclusion of 37 RCTs involving 2430 patients. The literature identification process is depicted in Fig. 1.

Characteristics of the included studies

The initial database searches identified 2,560 potentially relevant studies. From this pool, 1,627 duplicates were removed. Following this, a title and abstract screening process excluded 364 irrelevant records. A full-text assessment then eliminated 115 studies due to inaccessible data, resulting in 37 studies that met the inclusion criteria for NMA across fifteen therapeutic regimens, as summarized in Table 1 and Supplementary Table 5.The basic characteristics of the 37 published randomized controlled trials included in this study. These studies encompassed ten treatment models: LLLT ($n=21$), Chewing ($n=4$), Vibration ($n=4$), TEAS ($n=2$), Structured call ($n=2$), Ibuprofen postoperative ($n=6$), Ibuprofen preemptive ($n=4$), Ibuprofen preemptive postoperative ($n=1$), Etoricoxib preemptive postoperative ($n=1$), Acetaminophen preemptive ($n=2$), Aspirin preemptive postoperative ($n=1$), Naproxen preemptive postoperative ($n=1$), Naproxen preemptive ($n=1$), Acetaminophen postoperative ($n=1$), and Acetaminophen preemptive postoperative ($n=1$). Thirty-six studies utilized the Visual Analog Scale (VAS) to assess pain at 24 h following orthodontic treatment, while thirty studies employed the VAS for pain evaluation at 48 h post-treatment.

Results of the risk of bias

Two researchers (X.D and H.Q) independently assessed the risk of bias using the Cochrane Risk of Bias 2.0 tool [16]. Risk-of-bias summary visualizations were generated using the RoB 2.0 framework [17], identifying 4 studies high risk, 14 at some concerns, and 17 low risk studies in Fig. 2. and Fig. 3.

Risk of bias was assessed using the Cochrane RoB 2.0 tool, which evaluates potential bias across the following domains: bias arising from the randomization process, bias due to deviations from intended interventions, bias due to missing outcome data, bias in measurement of the outcome, and bias in selection of the reported result. Studies with a high risk of bias were excluded in the sensitivity analysis to test the stability of the results and ensure the robustness of the conclusions.

To assess the impact of study quality, a sensitivity analysis was performed by sequentially excluding studies with high risk of bias from the dataset. The results of this analysis showed that excluding high-risk studies did not alter the overall conclusions of the network meta-analysis.

Certainty of evidence (GRADE-NMA)

The confidence in the network meta-analysis estimates, as assessed using the GRADE-NMA framework, ranged from very low to high across interventions. Higher confidence ratings were mainly observed for pharmacological pre-post regimens and low-level laser therapy, whereas several non-pharmacological interventions

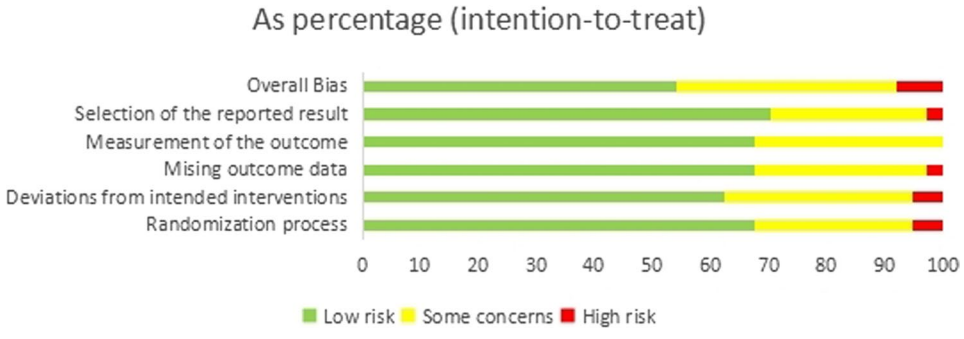


Fig. 1 Flow chart of study selection

Table 1 Characteristics of included studies

Study	Country	Intervention	Sample size	Outcome measures(VAS)	
				24 H	48 H
Al-Hanbali2024 [18]	Syrian	A: LLLT B: Vibration E: Placebo	A = 48 B = 48 E = 49	A:12.70 ± 9.50 B:11.35 ± 9.60 E:57.91 ± 27.20	A:2.18 ± 4.60 B:2.50 ± 5.25 E:47.91 ± 25.17
Mirhashemi 2021 [19]	Iran	A: LLLT E: Placebo	A = 26 E = 26	A:72.2 ± 19.5 E:79.2 ± 18.5	
Martins2019 [20]	Brazil	A: LLLT E: Placebo	A = 62 E = 62	A:16 ± 18 E:18 ± 20	A:8 ± 14 E:11 ± 14
Qamruddin2017 [21]	Pakistan	A: LLLT E: Placebo	A = 20 E = 20	A:14 ± 8.2 E:22 ± 4.1	A:10 ± 6.4 E:14 ± 10.4
Kim2013 [22]	Korea	A: LLLT E: Placebo	A = 28 E = 30	A:26.64 ± 6.28 E:50.47 ± 5.62	A:26.59 ± 6.28 E:47.11 ± 5.63
Prasad2019 [23]	India	A: LLLT E: Placebo	A = 10 E = 10	A:12 ± 11.7 E:25 ± 9.7	A:6 ± 8.4 E:12 ± 10.3
Stein2015 [24]	Germany	A: LLLT E: Placebo	A = 20 E = 20	A:20 ± 18 E:46 ± 34	A:20 ± 20 E:32 ± 26
Almallah Hasan2020 [25]	Syrian	A: LLLT E: Placebo	A = 18 E = 18	A = 22.61 ± 28.16 E = 33.28 ± 31.7	A:19.28 ± 24.59 E:31.22 ± 29.6
Eslamian2014 [26]	Iran	A: LLLT E: Placebo	A = 74 E = 74	A = 10.4 ± 17.2 E=:14.2 ± 22.5	
Hasan2020 [27]	Syrian	A: LLLT E: Placebo	A = 13 E = 13	A:14.92 ± 12.48 E:22 ± 24.28	A:10.23 ± 9.98 E:15.23 ± 21.01
Qamruddin2018 [28]	Pakistan	A: LLLT E: Placebo	A = 42 E = 42	A:20 ± 16 E:35 ± 26	A:33 ± 29 E:52 ± 31
Almallah2016 [29]	Syrian	A: LLLT E: Placebo	A = 18 E = 18	A:27.22 ± 20.99 E:41.78 ± 24	A:20.11 ± 21.85 E:30.28 ± 23.89
Ren2020 [30]	China	A: LLLT E: Placebo	A = 27 E = 27	A:8.1 ± 13.1 E:15.0 ± 4.0	A:4.2 ± 8.9 E:6.9 ± 1.0
Qamruddin2016 [31]	Pakistan	A: LLLT E: Placebo	A = 88 E = 88	A:18.6 ± 25.0 E:50.4 ± 32.4	A:15.4 ± 23.7 E:45.4 ± 32.7
Guram2018 [32]	India	A: LLLT E: Placebo	A = 20 E = 20	A:18.8 ± 22.1 E:51.6 ± 48.6	A:12.1 ± 31.2 E:22.3 ± 28.9
Ghaffar2022 [33]	Egypt	A: LLLT E: Placebo	A = 15 E = 15	A:30 ± 20 E:20 ± 9	A:39 ± 20 E:30 ± 16
Celebi2021 [34]	Turkey	A: LLLT B: Chewing E: Placebo	A = 21 B = 21 E = 21	A:48.0 ± 21.2 B:51.6 ± 20.8 E:53.4 ± 23.5	A:42.0 ± 23.7 B:42.7 ± 19.3 E:44.8 ± 24.6
Azizi2024 [35]	Iran	A: LLLT B: Chewing C: Ibuprofen_Postoperative E: Placebo	A = 27 B = 29 C = 27	A:25.9 ± 25.8 B:24.3 ± 29.9 E:20.4 ± 24.2	A:23.7 ± 28.1 B:17.4 ± 24.4 E:13.9 ± 17.9
Ahrari2024 [36]	Iran	A: LLLT B: TEAS C: Ibuprofen_Postoperative E: Placebo	A = 15 B = 15 E = 15	A:32.7 ± 25.2 B:38 ± 28.6 C:39.3 ± 25.5 E:31.3 ± 28.5	A:42.7 ± 28.4 B:39.3 ± 24.6 C:19 ± 16.4 E:32 ± 29.3
Owayda2022 [37]	Syrian	A: LLLT E: Placebo	A = 18 E = 18	A:37.7 ± 27.5 E:52.2 ± 35.1	A:33.3 ± 22.2 E:52.2 ± 25.1
Kaya2021 [38]	Turkey	A: LLLT B: Ibuprofen_Preemptive E: Placebo	A = 20 B = 20 E = 20	A:25 ± 15.2 B:20 ± 10.2 E:32.5 ± 18.8	
Çelebi2022 [39]	Turkey	A: Chewing B: Vibration E: Placebo	A = 19 B = 19 E = 19	A:52.1 ± 24.2 B:40.3 ± 29.5 E:52.6 ± 21.1	A:42.8 ± 20.8 B:32.5 ± 24.9 E:44.0 ± 22.3
Farzanegan2012 [40]	Iran	A: Chewing B: Ibuprofen_Postoperative E: Placebo	A = 10 B = 10 E = 10	A:34.7 ± 38.3 B:44.5 ± 20.7 E:74.7 ± 27.3	A:38.0 ± 33.9 B:47.6 ± 34.2 E:66.4 ± 31.1
Katchooi2018 [41]	American	A: Vibration E: Placebo	A = 13 E = 13	A:34 ± 24 E:47 ± 31	A:33 ± 19 E:43 ± 28

Table 1 (continued)

Study	Country	Intervention	Sample size	Outcome measures(VAS)	
				24 H	48 H
Alansari2018 [42]	American	A: Vibration E: Placebo	A = 14 E = 13	A:33.9 ± 13.5 E:41.9 ± 7.1	
Miles2012 [43]	Australia	A: Vibration E: Placebo	A = 30 E = 28	A:47.6 ± 24.5 E:41.5 ± 27.2	
Salmassian2009 [44]	American	A: Ibuprofen_Postoperative B: Acetaminophen_Postoperative E: Placebo	A = 19 B = 21 E = 20	A:36.3 ± 30.1 B:35.9 ± 27.7 E:57.91 ± 27.20	A:20.5 ± 25.3 B:17 ± 18.5 E:27.2 ± 21.4
Elshehaby2025 [45]	Egypt	A: Ibuprofen_Postoperative E: Placebo	A = 25 E = 21	A:32.8 ± 19.7 E:42.4 ± 2	
Kohli2011 [46]	India	A: Ibuprofen_Preemptive E: Placebo	A = 30 E = 30	A:53.2 ± 18.7 E:66.6 ± 21.9	A:48.7 ± 21.9 E:60.1 ± 30.1
Polat2005 [47]	Turkey	A: Ibuprofen_Preemptive B: Naproxen_Preemptive E: Placebo	A = 20 E = 20	A:54.6 ± 38.2 B:34.1 ± 32.7 E:59.4 ± 31.2	A:50.1 ± 31 B:36 ± 31.6 E:42.3 ± 28
Najaf2015 [48]	Iran	A: Ibuprofen_Preemptive B: Acetaminophen_Preemptive E: Placebo	A = 76 E = 76	A:17.8 ± 24.3 B:14.8 ± 19.1	A:17.1 ± 22.4 B:12 ± 20.1
Nik2016 [49]	Iran	A: Ibuprofen_Preemptive B: Acetaminophen_Preemptive E: Placebo	A = 29 B = 32 E = 28	A:25.27 ± 17.1 B:22.83 ± 18.73 E:58.26 ± 26.95	
Polat2005 [50]	Turkey	A: Ibuprofen_Preemptive_Postoperative B: Acetaminophen_Preemptive_Postoperative C: Naproxen_Preemptive_Postoperative D: Aspirin_Preemptive_Postoperative E: Placebo	A = 25 B = 25 C = 25 D = 25 E = 25	A:24.5 ± 3.26 B:13.1 ± 24.7 C:10.5 ± 26.2 D:18.2 ± 32.3 E:59.4 ± 31.2	A:23.5 ± 27.8 B:16.6 ± 24.9 C;9.6 ± 22.7 D:14 ± 22.1 E:42.3 ± 28
Gupta2014 [51]	India	A: Acetaminophen_Preemptive_Postoperative B: Etoricoxib_Preemptive_Postoperative E: Placebo	A = 15 B = 15 E = 15	A:32 ± 8 B:11 ± 1 E:64 ± 11	A:26 ± 8 B:11 ± 1 E:55 ± 11
Kaur2019 [51]	India	A: Structured call E: Placebo	A = 34 E = 34		A:43.8 ± 15.38 E:37.1 ± 14.47
Bartlett2005 [52]	American	A: Structured call E: Placebo	A = 49 E = 54	A:34.62 ± 2.84 E:45.79 ± 2.83	A:26.71 ± 2.84 E:14 ± 10.4
Roth [53]	American	A: TEAS E: Placebo	A = 9 E = 9	A:4.77 ± 6.96 E:14.55 ± 6.89	A:5.55 ± 9.3 E:16.44 ± 8.24

A, B,C, D, experimental group; E, control group; VAS, visual analog scale; LLLT, low-level laser therapy; TEAS, transcutaneous electrical acupoint stimulation

were supported by low or very low confidence evidence. Downgrading of the certainty of evidence was primarily driven by risk of bias and imprecision. Detailed GRADE-NMA assessments for pain outcomes at 24 h and 48 h are presented in Supplementary Tables 6 and 7.

Network meta-analysis results

24HVAS

This network meta-analysis integrated 2,322 participants from 36 randomized controlled trials (RCTs) assessing 15 interventions: low-level laser therapy (LLLT); vibration therapy; chewing therapy; transcutaneous electrical acupoint stimulation (TEAS); structured telephone follow-up; various pharmacological regimens including naproxen preemptive, naproxen postoperative, naproxen preemptive–postoperative, ibuprofen preemptive, ibuprofen postoperative, ibuprofen preemptive–postoperative, acetaminophen preemptive, acetaminophen postoperative, and acetaminophen preemptive–postoperative administration; as well as etoricoxib and aspirin preemptive–postoperative administration.

The network topology comprised 24 closed loops (Supplementary Fig. 1A), and both global and node-splitting analyses confirmed network consistency ($P > 0.05$). Forest plots versus placebo (Supplementary Fig. 2A) and Table 2 demonstrated statistically significant 24-hour pain reduction for eight interventions: acetaminophen preemptive (MD: −23.64; 95% CrI: −39.01 to − 8.40), acetaminophen preemptive–postoperative (MD: −27.41; 95% CrI: −46.33 to − 7.61), ibuprofen preemptive–postoperative (MD: −25.12; 95% CrI: −42.45 to − 7.23), naproxen preemptive (MD: −33.55; 95% CrI: −53.28 to − 13.45), naproxen postoperative (MD: −27.39; 95% CrI: −48.04 to − 6.91), etoricoxib preemptive–postoperative (MD: −38.64; 95% CrI: −58.32 to − 18.40), structured telephone follow-up (MD: −19.26; 95% CrI: −36.02 to − 2.47), and LLLT (MD: −18.92; 95% CrI: −35.04 to − 2.56).

Based on SUCRA (Supplementary Fig. 4) rankings, etoricoxib preemptive–postoperative (99.35%) and

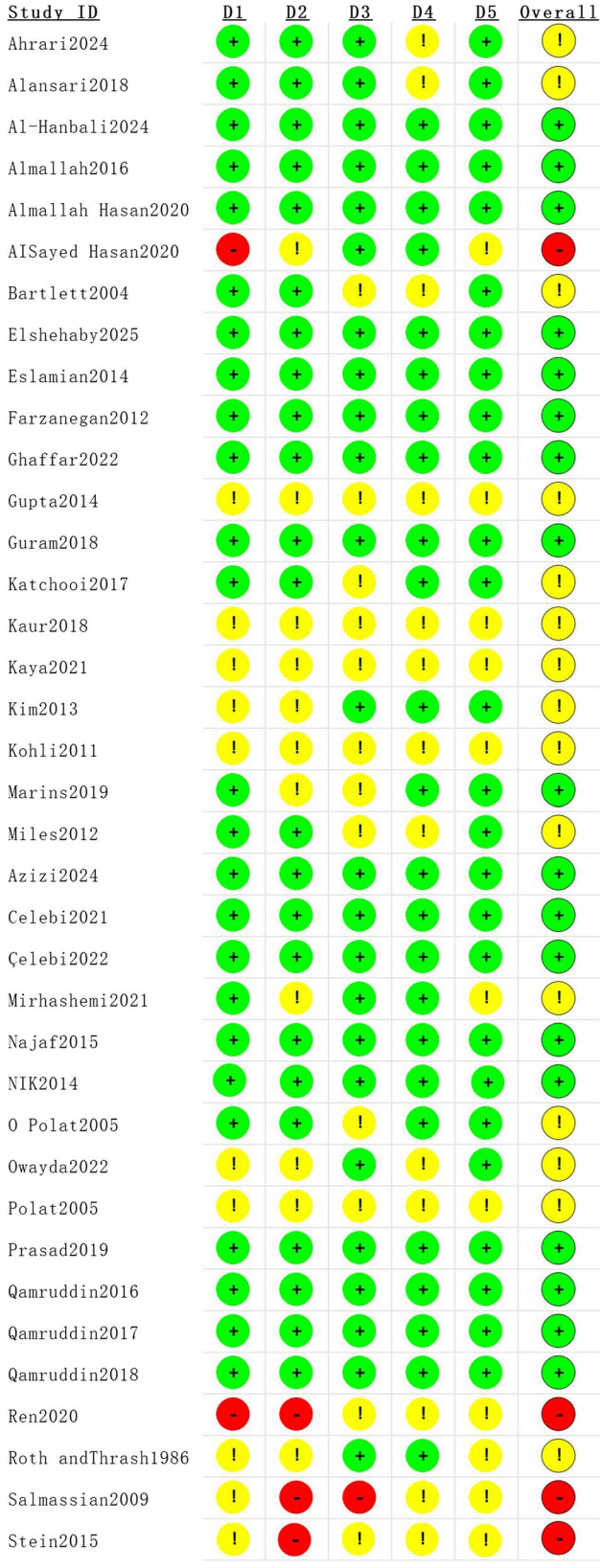


Fig. 2 Risk of bias graph

naproxen preemptive (91.64%) were the most effective interventions for 24-hour pain reduction, followed by naproxen postoperative (86.49%) and acetaminophen preemptive–postoperative (82.34%). Non-pharmacological interventions such as LLLT and structured telephone follow-up showed moderate benefits, ranking above vibration therapy and chewing therapy.

48HVAS

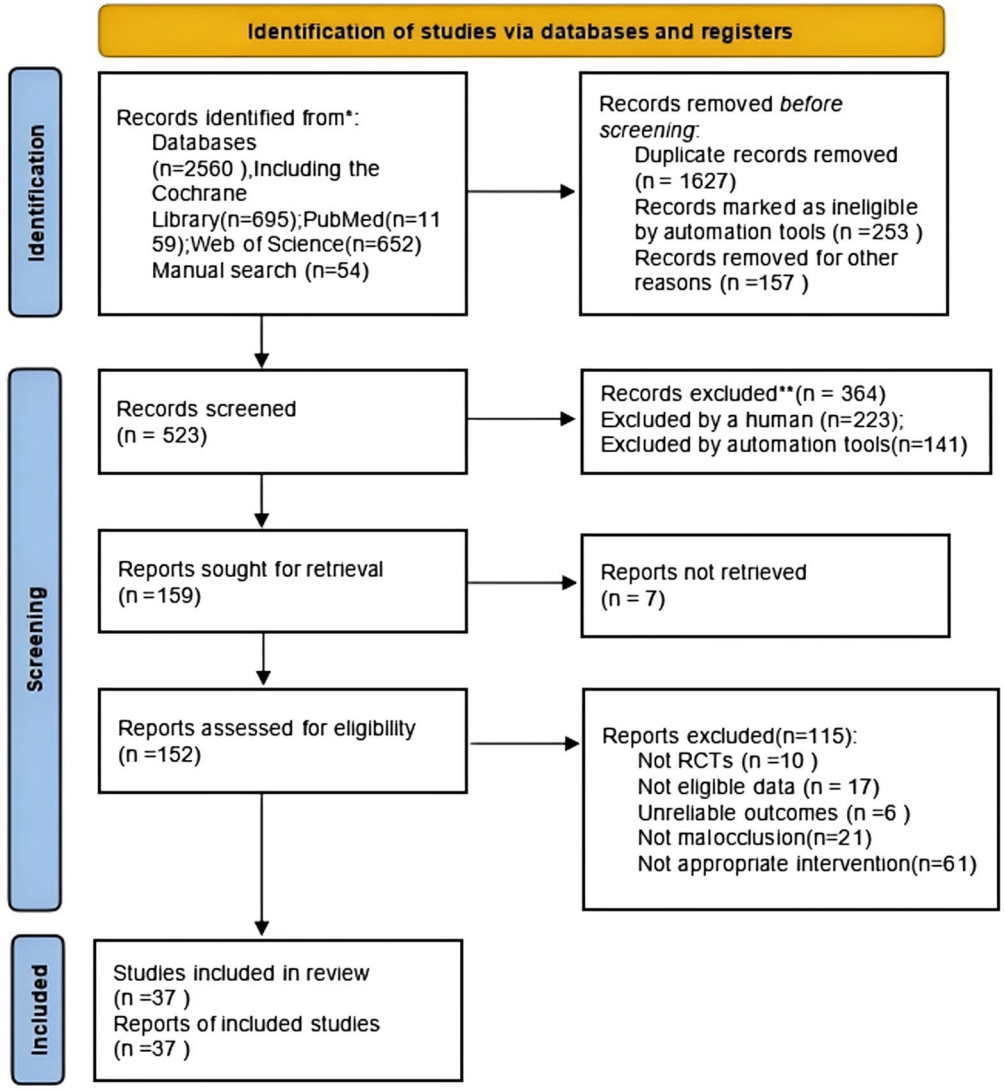
This network meta-analysis incorporated 1,995 participants evaluating 16 interventions: low-level laser therapy (LLLT); vibration therapy; chewing therapy; transcutaneous electrical acupoint stimulation (TEAS); structured telephone follow-up; naproxen preemptive, postoperative, and preemptive–postoperative administration; ibuprofen preemptive, postoperative, and preemptive–postoperative administration; acetaminophen preemptive, postoperative, and preemptive–postoperative administration; etoricoxib preemptive–postoperative administration; aspirin preemptive–postoperative administration; and attention-based telephone follow-up.

The network topology included 25 closed loops (Supplementary Fig. 1B), with global and node-splitting analyses confirming consistency ($P > 0.05$). Forest plots versus placebo (Supplementary Fig. 2B) and Table 3 revealed statistically significant 48-hour pain reduction for nine interventions: acetaminophen preemptive–postoperative (MD: −25.98; 95% CrI: −44.52 to −6.84), ibuprofen preemptive–postoperative (MD: −22.77; 95% CrI: −40.12 to −5.01), naproxen preemptive–postoperative (MD: −31.45; 95% CrI: −50.02 to −12.38), naproxen postoperative (MD: −25.89; 95% CrI: −45.18 to −6.67), etoricoxib preemptive–postoperative (MD: −36.81; 95% CrI: −57.02 to −16.22), structured telephone follow-up (MD: −18.36; 95% CrI: −33.87 to −2.28), vibration therapy (MD: −17.52; 95% CrI: −31.44 to −3.15), TEAS (MD: −16.08; 95% CrI: −30.12 to −1.85), and LLLT (MD: −15.49; 95% CrI: −28.44 to −2.35).

According to SUCRA (Supplementary Fig. 5) results, etoricoxib preemptive–postoperative (90.45%) and naproxen preemptive–postoperative (78.78%) remained the most effective for 48-hour pain reduction. Structured telephone follow-up, vibration therapy, and LLLT also demonstrated favorable outcomes compared with placebo, confirming their moderate analgesic effects at 48 h post-intervention.

Publication bias

Funnel plots were created to assess publication bias, with the 24-hour and 48-hour visualizations shown in Supplementary Figs. 3. The analysis of visual asymmetry revealed a largely symmetrical inverted funnel distribution, where most effect estimates were concentrated near the apex, with few outliers beyond the pseudo



*Consider, if feasible to do so, reporting the number of records identified from each database or register searched (rather than the total number across all databases/registers).

**If automation tools were used, indicate how many records were excluded by a human and how many were excluded by automation tools.

Fig. 3 Risk of bias summary of the included studies

95% confidence limits. These findings suggest a low risk of publication bias and support the robustness of the results.

Discussion

The findings from this network meta-analysis (NMA) indicate that low-level laser therapy(LLLT), vibration therapy, chewing therapy, transcutaneous electrical acupoint stimulation (TEAS), structured telephone calls, naproxen, aspirin, ibuprofen, acetaminophen, and etoricoxib are all effective in reducing post-orthodontic pain, as assessed by VAS. Moreover, pharmacological treatments are superior to non-pharmacological methods, with etoricoxib and naproxen ranking as the most effective options in our analysis. Among non-pharmacological interventions, TEAS showed the best results at 24 h post-treatment, while structured telephone calls were most effective at 48 h.

The reductions in VAS scores observed in this analysis were clinically meaningful. A 10–20 mm change on the 100-mm VAS is generally regarded as the minimal clinically important difference (MCID) for acute pain [54].

Table 2 Pairwise comparisons (row intervention vs. column intervention) in network meta-analysis for 24 h VAS

A	A															
B	16.64 (-10.19, 43.54)	B														
C	30.48 (3.92, 57.24)	13.79 (-7.51, 35.17)	C													
D	-11.74 (-43.42, 19.95)	-28.48 (-55.72, -1.05)	-42.29 (-64.49, -20.05)	D												
E	-0.081 (-24.28, 24.26)	-16.73 (-35.42, 2.09)	-30.56 (-49.15, -11.98)	11.71 (-13.61, 37.02)	E											
F	48.93 (20.65, 77.22)	32.23 (8.87, 55.69)	18.48 (0.82, 35.73)	60.71 (34.13, 87.08)	49.03 (28.01, 69.78)	F										
G	2.13 (-20.17, 24.13)	-14.55 (-32.85, 3.55)	-28.33 (-46.58, -10.53)	13.91 (-11.01, 38.60)	2.22 (-11.15, 15.30)	-46.79 (-67.25, -26.48)	G									
H	10.42 (-14.02, 34.67)	-6.25 (-19.72, 6.91)	-20.04 (-38.40, -2.11)	22.20 (-3.04, 46.95)	10.48 (-4.76, 25.39)	-38.49 (-59.17,-18.09)	8.30 (-6.04, 22.55)	H								
I	23.13 (-6.19, 52.31)	6.42 (-18.06, 31.04)	-7.36 (-25.97, 11.26)	34.90 (13.21, 56.58)	23.18 (0.98, 45.55)	-25.81 (-49.28, -2.08)	20.97 (-0.51, 42.86)	12.66 (-8.85, 34.81)	I							
J	2.89 (-19.52, 25.25)	-13.79 (-29.59, 1.93)	-27.60 (-43.40, -12.05)	14.68 (-8.62, 37.68)	2.96 (-8.50, 14.13)	-46.04 (-64.41, -27.71)	0.75 (-9.621, 11.17)	-7.55 (-18.35, 3.43)	-20.22 (-40.10, -0.63)	J						
K	23.79 (-8.86, 56.42)	7.037 (-20.14, 34.36)	-6.69 (-35.24, 21.52)	35.52 (2.39, 68.53)	23.82 (-2.69, 50.26)	-25.16 (-54.90, 4.66)	21.61 (-4.41, 47.73)	13.35 (-11.01, 37.80)	0.64 (-30.16, 31.25)	20.89 (-3.48, 45.43)	K					
L	37.15 (6.16, 68.05)	20.42 (-6.21, 47.14)	6.64 (-14.58, 27.98)	48.95 (24.91, 72.85)	37.20 (12.65, 61.75)	-11.80 (-37.56, 14.02)	34.99 (11.09, 59.21)	26.71 (2.84, 50.96)	14.01 (-6.80, 34.68)	34.24 (11.89, 56.64)	13.37 (-19.01, 45.95)	L				
M	-6.97 (-29.09, 14.99)	-23.64 (-39.01, -8.40)	-37.45 (-52.59, -22.60)	4.81 (-18.08, 27.40)	-6.88 (-18.14, 4.06)	-55.90 (-73.75, -38.22)	-9.10 (-19.06, 0.87)	-17.40 (-27.66, -7.05)	-30.06 (-49.48, -11.02)	-9.85 (-14.48, -5.28)	-30.74 (-54.94, -6.73)	-44.09 (-66.05, -22.31)	M			
N	4.20 (-24.22, 32.60)	-12.45 (-36.10, 10.99)	-26.26 (-49.83, -3.09)	16.01 (-13.06, 44.91)	4.29 (-17.13, 25.26)	-44.69 (-70.08, -19.50)	2.08 (-18.39, 22.64)	-6.22 (-26.93, 14.61)	-18.91 (-45.31, 7.31)	1.31 (-17.29, 19.88)	-19.56 (-49.59, 10.33)	-32.94 (-61.36, -4.84)	11.16 (-6.83,29.26)	N		
O	-0.01 (-26.23, 26.02)	-16.64 (-37.93, 4.26)	-30.41 (-51.53, -9.96)	11.81 (-15.48, 38.48)	0.13 (-18.20, 17.76)	-48.89 (-71.94, -26.25)	-2.14 (-18.84, 14.51)	-10.38 (-28.15, 7.23)	-23.04 (-47.46, 0.57)	-2.85 (-17.90, 11.90)	-23.72 (-52.04, 4.23)	-37.07 (-63.57, -11.19)	7.01 (-7.55, 21.31)	-4.14 (-27.43, 18.73)	O	
P	2.28 (-21.68, 26.14)	-14.40 (-32.50, 3.47)	-28.19 (-46.06, -10.59)	14.08 (-10.85, 38.56)	2.36 (-11.56, 15.87)	-46.62 (-66.78, -26.55)	0.13 (-13.34, 13.67)	-8.14 (-22.06, 5.79)	-20.82 (-42.54, 0.40)	-0.58 (-10.67, 9.32)	-21.46 (-47.40, 4.26)	-34.87 (-58.74, -11.07)	9.26 (-0.24, 18.69)	-1.93 (-22.25, 18.44)	2.23 (-14.77, 19.51)	P

A:Acetaminophen postoperative; B:Acetaminophen preemptive; C:Acetaminophen preemptive postoperative; D:Aspirin preemptive postoperative; E:Chewing; F:Etoricoxib preemptive postoperative; G:Ibuprofen postoperative; H:Ibuprofen preemptive; I:Ibuprofen preemptive postoperative; J:LLLT; K:Naproxen preemptive; L:Naproxen preemptive postoperative; M:Placebo; N:Structured phone call; O:TEAS; P:Vibration. Values represent weighted mean differences (WMDs) with 95% credible intervals.Negative values indicate a reduction in pain intensity favoring the intervention listed in the row compared with the intervention listed in the column

Several interventions, such as etoricoxib and naproxen, exceeded this threshold, indicating tangible improvements in patient comfort.

Post-orthodontic pain generally begins around 6 h after the procedure [55], reaches its peak between 24 and 48 h, although there is no definitive agreement on this timeline, and typically subsides within a week [56]. Consequently, this study exclusively incorporated literature reporting VAS measurements at 24- and 48-hour intervals.

In terms of pharmacological treatment, the timing of drug administration was further differentiated. The drugs were categorized into three phases: preoperative, postoperative, and combined preoperative and postoperative administration. Clinically, these findings support an individualized approach to pain control. Etoricoxib and naproxen appear to be the most effective NSAIDs for short-term orthodontic pain relief, while non-pharmacological methods—such as low-level laser therapy and vibration therapy—offer feasible, non-invasive options for patients seeking drug-free management.This study sought to deliver more accurate estimates, even in scenarios where multiple intervention techniques exist and there are few studies for each comparison. The results suggest that combined preoperative and postoperative administration of etoricoxib yields the best effect, which is consistent with the findings of a review analysis by Satpal [57].Etoricoxib is a potent and selective COX-2 inhibitor within the NSAID class, exhibiting significant anti-inflammatory and analgesic properties [58]. Additionally, combined preoperative and postoperative administration can extend the metabolism time of the drug in the body, leading to better analgesic effects. For example, etoricoxib has a plasma half-life of 22 h, and naproxen has a plasma half-life of 15 h. Therefore, the combined preoperative and postoperative administration enhances the effectiveness of these drugs [58]. On the other hand, analgesics that have shorter plasma half-lives, like aspirin (around 6 h), show reduced analgesic efficacy, which can be attributed to their primary antipyretic effects rather than their analgesic actions through central mechanisms.

Ibuprofen, a common nonsteroidal anti-inflammatory drug (NSAID), primarily exerts its effects through inhibition of cyclooxygenase (COX) isoforms, thereby suppressing prostaglandin (PG) synthesis [59].These drugs are known as non-selective COX inhibitors because they block both COX-1 and COX-2. Therefore, it is clear that pharmacological therapy stands out as the most effective approach for managing post-orthodontic pain. NSAIDs alleviate postoperative pain by inhibiting the production of prostaglandins (PGs) [55, 60].PGs serve as critical mediators in orthodontic therapy, where they orchestrate inflammatory responses within the periodontal ligament (PDL) following force application, facilitate tooth movement, and regulate bone resorption-apposition coupling [61].Consequently, emerging evidence indicates that while NSAIDs effectively alleviate postoperative pain, they concurrently slow down orthodontic tooth movement [62].In contrast, there have been no reports of acetaminophen having adverse effects on tooth movement [1, 63].

Acetaminophen, a widely used over-the-counter (OTC) analgesic, achieves its therapeutic effects by centrally inhibiting cyclooxygenase (COX) activity, with a primary focus on COX isoforms found in the cerebral cortex and cardiac tissue, often referred to as the putative “COX-3” pathway. This mechanism confers both antipyretic and analgesic properties, with demonstrated significant efficacy in managing post-orthodontic pain [64].

TEAS (Transcutaneous Electrical Acupoint Stimulation) can be tailored to the patient’s condition by selecting specific acupoints and adjusting parameters accordingly. TEAS offers specific benefits in alleviating pain-related symptoms and improving overall quality of life by delivering microcurrents to targeted acupoints through the skin [65].In this NMA, TEAS demonstrated a high likelihood of being the most effective non-pharmacological intervention at the 24-hour time point, with excellent efficacy also observed at 48 h. This finding aligns closely with the views presented in Li’s study [66]. The reason could be attributed to the ability of low-frequency TEAS stimulation to encourage the sustained release of endorphins [67], which, when activated, can offer more effective and longer-lasting analgesic effects.

Structured telephone calls and attention calls are both forms of verbal behavior modification (VBM), differing only in the content of the conversation. Research has shown that pain is a subjective sensation influenced by pathological changes as well as various biological and physiological factors [11].In Maggirias’s study, psychological factors were found to exert a pronounced effect on pain perception: approximately 4%–20% of patients exhibit symptoms of dental anxiety, which in turn lowers their pain threshold [68].Therefore, VBM interventions exemplified by structured telephone calls can mitigate anxiety via psychological pathways, thereby more effectively assisting patients in alleviating pain.

The principle of mechanical vibration therapy is that the temporary displacement of teeth can loosen compressed areas of the periodontal tissues, including nerve fibers and occluded blood vessels, thereby facilitating better blood flow [69]. As the duration of vibration increases, there is a greater reduction in interleukin-1 secretion, which correlates with more effective pain relief. This may explain why vibration therapy at 48 h produces superior analgesic effects compared to 24 h. In addition, Furthermore, animal studies suggest that high-frequency

Table 3 Pairwise comparisons (row intervention vs column intervention) in network meta-analysis for 48 h VAS

A	A														
B	-7.71 (-56.15, 40.42)	B													
B	12.08 (-24.50, 48.69)	19.75 (-24.88, 64.74)	C												
D	13.72 (-28.10, 55.87)	21.44 (-27.90, 70.77)	1.71 (-28.47, 31.93)	D											
E	6.54 (-45.36, 58.56)	14.27 (-43.61, 72.17)	-5.58 (-53.71, 42.89)	-7.30 (-59.70, 45.62)	E										
F	-2.33 (-34.34, 29.75)	5.41 (-36.29, 47.36)	-14.41 (-41.61, 13.01)	-16.11 (-50.31, 18.19)	-8.85 (-54.51, 36.70)	F									
G	27.82 (-12.64, 68.58)	35.57 (-12.43, 83.79)	15.79 (-12.37, 44.11)	14.10 (-24.32, 52.75)	21.35 (-30.12, 72.58)	30.17 (-2.22, 62.62)	G								
H	2.21 (-26.96, 31.34)	9.98 (-31.70, 51.92)	-9.81 (-37.62, 17.60)	-11.46 (-46.02, 22.72)	-4.21 (-50.20, 41.32)	4.58 (-14.82, 23.66)	-25.61 (-58.41, 7.021)	H							
I	-12.79 (-50.37, 24.55)	-5.06 (-35.56, 25.43)	-24.87 (-57.81, 7.87)	-26.53 (-65.35, 11.98)	-19.32 (-68.53, 29.59)	-10.42 (-39.14, 17.96)	-40.64 (-77.92, -3.67)	-15.04 (-43.76, 13.65)	I						
J	4.28 (-38.29, 46.98)	11.99 (-37.51, 61.98)	-7.78 (-38.56, 23.20)	-9.43 (-42.58, 23.43)	-2.19 (-55.38, 50.52)	6.64 (-28.45, 41.63)	-23.58 (-62.46, 15.36)	2.04 (-33.01, 37.12)	17.02 (-22.01, 56.53)	J					
K	-1.50 (-31.25, 27.99)	6.27 (-33.27, 45.50)	-13.60 (-37.33, 9.99)	-15.25 (-46.73, 15.94)	-7.98 (-51.71, 35.29)	0.83 (-15.41, 16.72)	-29.35 (-58.92, -0.11)	-3.74 (-20.22, 12.62)	11.28 (-13.76, 36.22)	-5.73 (-37.89, 26.19)	K				
L	-3.99 (-47.67, 39.75)	3.79 (-41.25, 48.74)	-15.97 (-55.94, 23.75)	-17.71 (-62.40, 26.90)	-10.45 (-64.34, 43.20)	-1.57 (-38.16, 34.51)	-31.79 (-75.26, 11.57)	-6.18 (-42.56, 30.48)	8.88 (-24.14, 41.79)	-8.24 (-53.46, 36.79)	-2.46 (-36.02, 31.26)	L			
M	18.15 (-23.88, 60.13)	25.91 (-23.28, 75.27)	6.15 (-24.12, 36.42)	4.45 (-27.97, 36.53)	11.67 (-41.01, 64.28)	20.50 (-13.96, 54.68)	-9.68 (-48.241, 28.85)	15.94 (-18.56, 50.46)	30.91 (-7.56, 69.93)	13.88 (-19.05, 46.98)	19.64 (-11.65, 51.10)	22.07 (-22.70, 67.03)	M		
N	-15.36 (-44.36, 13.59)	-7.60 (-46.22, 31.17)	-27.43 (-49.95, -5.05)	-29.11 (-59.53, 1.33)	-21.86 (-64.80, 20.93)	-13.01 (-28.75, 2.58)	-43.21 (-71.67, -14.84)	-17.61 (-33.53, -1.49)	-2.55 (-26.29, 21.39)	-19.63 (-50.71, 11.47)	-13.85 (-21.26, -6.21)	-11.39 (-44.04, 21.42)	-33.50 (-63.91, -2.96)	N	
O	-0.20 (-41.86, 41.70)	7.63 (-41.27, 56.37)	-12.21 (-49.61, 25.09)	-13.88 (-56.49, 28.77)	-6.71 (-37.43, 23.84)	2.18 (-31.75, 35.73)	-28.03 (-69.35, 13.22)	-2.43 (-36.29, 31.65)	12.67 (-25.57, 50.94)	-4.47 (-47.57, 38.65)	1.34 (-29.43, 32.23)	3.80 (-40.56, 48.24)	-18.32 (-61.02, 24.35)	15.20 (-14.82, 45.04)	O

Table 3 (continued)

P	-9.24 (-44.43, 25.66)	-1.42 (-45.58, 42.48)	-21.24 (-52.30, 9.47)	-22.94 (-60.36, 13.96)	-15.79 (-63.64, 31.91)	-6.86 (-32.84, 18.56)	-37.07 (-72.73, -1.82)	-11.48 (-35.78, 12.77)	3.63 (-28.28, 35.25)	-13.46 (-51.42, 24.07)	-7.70 (-29.59, 14.16)	-5.24 (-44.34, 33.54)	-27.35 (-64.76, 9.77)	6.12 (-15.14, 27.29)	-9.05 (-45.79, 27.60)	P	
Q	5.15 (-28.60, 38.60)	12.89 (-29.53, 55.25)	-6.96 (-35.39, 21.37)	-8.64 (-43.82, 26.30)	-1.42 (-47.68, 44.73)	7.47 (-13.91, 28.54)	-22.70 (-56.20, 10.59)	2.89 (-20.08, 25.73)	17.99 (-11.41, 47.40)	0.83 (-34.92, 36.37)	6.65 (-11.37, 24.73)	9.10 (-27.92, 46.06)	-13.04 (-48.16, 22.14)	20.51 (3.01 , 37.74)	5.29 (-29.37, 39.79)	14.36 (-12.72, 41.67)	Q

A:Acetaminophen postoperative; B:Acetaminophen preemptive; C:Acetaminophen preemptive postoperative; D:Aspirin preemptive postoperative E: Attention phone call; F:Chewing; G:Etoricoxib preemptive postoperative; H:Ibuprofen postoperative; I:Ibuprofen preemptive; J:Ibuprofen preemptive postoperative; K:LLLT; L:Naproxen preemptive; M:Naproxen preemptive postoperative; N:Placebo; O:Structured phone call; P:TEAS; Q:Vibration. Values represent weighted mean differences (WMDs) with 95% credible intervals.Negative values indicate a reduction in pain intensity favoring the intervention listed in the row compared with the intervention listed in the column

vibration may not only accelerate tooth movement but also lower the risk of root resorption [70].Additionally, it promotes the differentiation of osteoblasts, fibroblasts, and osteoclasts, enhancing bone-remodeling capacity and thereby speeding orthodontic tooth movement [71]. However, among the studies included in this analysis, differences in vibration device type, frequency, and duration significantly influenced VAS outcomes. Consequently, translating these potential benefits on treatment efficiency and safety from animal models to clinical practice requires further validation in human trials.

Among non-pharmacological interventions, low level laser therapy (LLLT) is the most extensively studied in the articles included here. LLLT reduces pain by inhibiting metabolites of arachidonic acid that interact with nociceptors, while also encouraging the release of endorphins, the body's natural pain-relieving peptides [72].Additionally, during orthodontic tooth movement, LLLT enhances physiological processes such as cellular metabolism, angiogenesis, cell proliferation, and fibronectin expression, thereby accelerating tooth movement and shortening overall treatment duration [73].However, substantial heterogeneity exists among the included laser therapy studies, primarily owing to variations in laser type, wavelength, irradiation time, power, and frequency. In this network meta-analysis, wavelengths were primarily grouped around 830 nm, all administered through diode lasers, with those in the 800–830 nm range showing the most significant analgesic effects [74]. It was observed that in the meta-analyses conducted by LI[62] and Satpa[52], LLLT demonstrated better pain control, which contrasts with the findings of the current study. This discrepancy occurs due to the fact that the two studies we included, Azizi 2024[32] and Ahrari 2024[33], utilized sham irradiation as the placebo control; in both cases, LLLT did not show a significant benefit over placebo. Those authors propose that patients' psychological expectations of laser treatment may predominantly drive their perceived analgesia. As a result, these trials were incorporated to provide a different perspective. Chewing therapy primarily involves chewing gum or bite blocks and operates on principles similar to vibration therapy. The mechanism operates through the loosening of tightly arranged periodontal ligament fibers that encase neural and vascular tissues, thereby alleviating inflammation and swelling and ultimately diminishing pain and discomfort [75].However, its effectiveness remains a subject of significant debate, with some orthodontists believing that chewing therapy may increase the likelihood of bracket fractures [76]. Although chewing therapy was shown to reduce postoperative pain in this meta-analysis, it ranked relatively low, and the number of included studies was limited. Further experimental data are needed to provide more conclusive evidence.

The applicability of these interventions depends on several factors, including patient preferences, availability of resources, and the clinical setting. For instance, NSAIDs are widely accessible and commonly used in clinical practice, making them highly applicable to a broad patient population. On the other hand, interventions like LLLT and vibration therapy may require specialized equipment, limiting their use to certain settings, such as orthodontic clinics equipped with these technologies. Further studies are needed to explore the cost-effectiveness and feasibility of implementing these treatments in various clinical environments.

Subgroup analyses by age and gender were considered but could not be performed due to insufficient stratified data and small subgroup sample sizes. Future studies should report demographic-specific pain outcomes to facilitate more detailed comparative analyses.

Evidence from the included trials suggests that etoricoxib 60–90 mg once daily and naproxen 500 mg twice daily, administered preemptively or within the first 24 h, achieved significant pain reduction with good tolerability. These regimens may serve as practical references for short-term analgesic use in clinical orthodontic care. Although pharmacological treatments tend to be more effective than non-pharmacological methods, the 2023 Clinical S3 guidelines highlight that systemic medication may have side effects, and long-term use should be avoided whenever possible [77].

Future studies should aim to standardize non-pharmacological intervention protocols—such as laser wavelength, energy density, vibration frequency, and timing—to enhance comparability across studies. Reporting outcomes stratified by age and gender is also encouraged to better understand demographic influences on pain perception. Therefore, we recommend non-pharmacological treatments for patients experiencing mild to moderate post-orthodontic pain and pharmacological treatments for those with severe pain to achieve optimal outcomes.

Conclusions

The results indicate that pharmacological interventions are the most effective measures for managing orthodontic pain, with etoricoxib demonstrating the greatest analgesic efficacy and emerging as the most potent pain reliever. Among non-pharmacological approaches, structured telephone calls, vibration therapy, and low-level laser therapy (LLLT) show promising clinical potential. However, more detailed data and optimized treatment parameters are required for further investigation.

Supplementary Information

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Supplementary Material 1.

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